

Geometry Quest

Every graded set, every week — 5 weeks, 25 two-page sheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

WEEK 5

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

What an Angle Is

Every answer is a number of degrees, or a count. Type digits only.

Warm-Up 8 ITEMS • RECALL

1 Right angles in a full turn

2 Right angles in a straight line

3 Perimeter of a 5 by 2 rectangle

4 Perimeter of a 4 by 4 square

5 Perimeter of a 8 by 3 rectangle

6 Perimeter of a 7 by 7 square

7 Perimeter of a 9 by 1 rectangle

8 Perimeter of a 2 by 2 square

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

9. Perimeter of a 14 by 6 rectangle

.....

10. A square of perimeter 24 — one side

.....

11. A rectangle of perimeter 26, one side 8 — the other

.....

12. Perimeter of a 20 by 11 rectangle

.....

13. A square of perimeter 64 — one side

.....

★

Challenge

NOT GRADED · REASON IT OUT

14. Degrees in a right angle
-
15. Degrees in a straight line
-
16. Degrees all the way round
-
17. Degrees in half a right angle
-
18. On a straight line: 130° and $?^\circ$ (180 - 130)
-
19. Round a point: 90° , 120° and $?^\circ$ (360 - 210)
-
20. Three angles of a triangle: 60° , 70° and $?^\circ$ (They add to 180)
-
21. An angle of 200° — how far past a straight line? (200 - 180)
-
22. Two equal angles on a straight line — each is $?^\circ$ (180 shared evenly)
-
23. Angles in a square, added up (Four right angles)
-
24. Turn 90° four times — total degrees (All the way round)
-
25. A rectangle of perimeter 24 with whole sides — how many different ones
-

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

!

Error journal

ONE SENTENCE PER MISTAKE

Perimeter & Area

Perimeter is the fence. Area is the grass. Read which one is being asked for.

Warm-Up 12 ITEMS • RECALL

1 Perimeter of a 5 by 3 rectangle

2 Area of a 5 by 3 rectangle

3 Perimeter of a square with side 6

4 Area of a square with side 6

5 Perimeter of a 10 by 2 rectangle

6 Area of a 10 by 2 rectangle

7 Area of a 5 by 2 rectangle

8 Area of a 4 by 4 square

9 Area of a 8 by 3 rectangle

10 Area of a 7 by 7 square

11 Area of a 9 by 1 rectangle

12 Area of a 6 by 5 rectangle

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Area 48, one side 6 — the other side ($48 \div 6$)

14. Perimeter 30, one side 9 — the other side ($9 + 9$ is 18, and 12 is left for two sides)

15. Area of a 12 by 7 rectangle (84 square units)

16. Perimeter of a 12 by 7 rectangle ($12 + 12 + 7 + 7$)

17. L-shape: a 6 by 4 block plus a 3 by 2 block — total area ($24 + 6$)

18. Area of a 14 by 6 rectangle

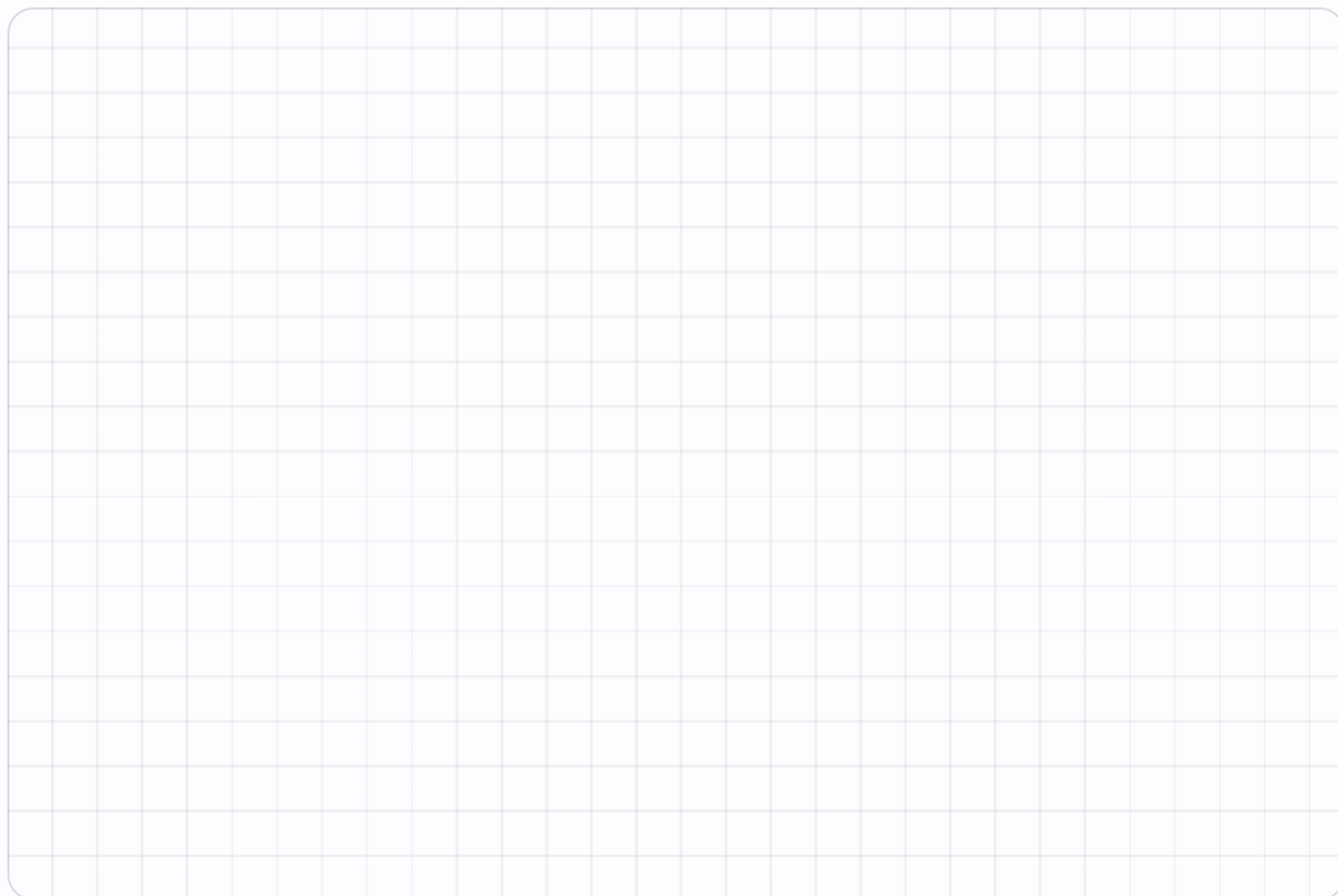
★ Challenge NOT GRADED • REASON IT OUT

23. Biggest area with perimeter 24 (whole sides) (The 6 by 6 square)

24. Smallest area with perimeter 24 (whole sides) (1 by 11)

25. Area of an L: a 7 by 7 square minus a 3 by 2 corner

26. Double both sides of a 3 by 6 rectangle — the new area

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

The Coordinate Plane

Across first, then up. (3, 5) is a different place from (5, 3).

Warm-Up 12 ITEMS • RECALL

1 In (3, 5), the across number

2 In (3, 5), the up number

3 In (7, 2), the across number

4 In (0, 4), the across number

5 In (6, 6), the up number

6 In (9, 1), the up number

7 A 1 by 9 rectangle — perimeter

8 That rectangle — area

9 A 5 by 5 square — perimeter

10 That square — area

11 A 2 by 8 rectangle — area

12 A 3 by 7 rectangle — area

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Distance from (2, 3) to (7, 3) (Only the across number changed)

14. Distance from (4, 1) to (4, 8) ($8 - 1$)

15. Rectangle at (1,1), (6,1), (6,4), (1,4) — its perimeter (5 wide and 3 tall)

16. That same rectangle — its area (5×3)

17. Start at (2,2), go 4 across and 3 up — the up number now ($2 + 3$)

18. With 20 m of fence, the biggest whole-number area

★ Challenge NOT GRADED • REASON IT OUT

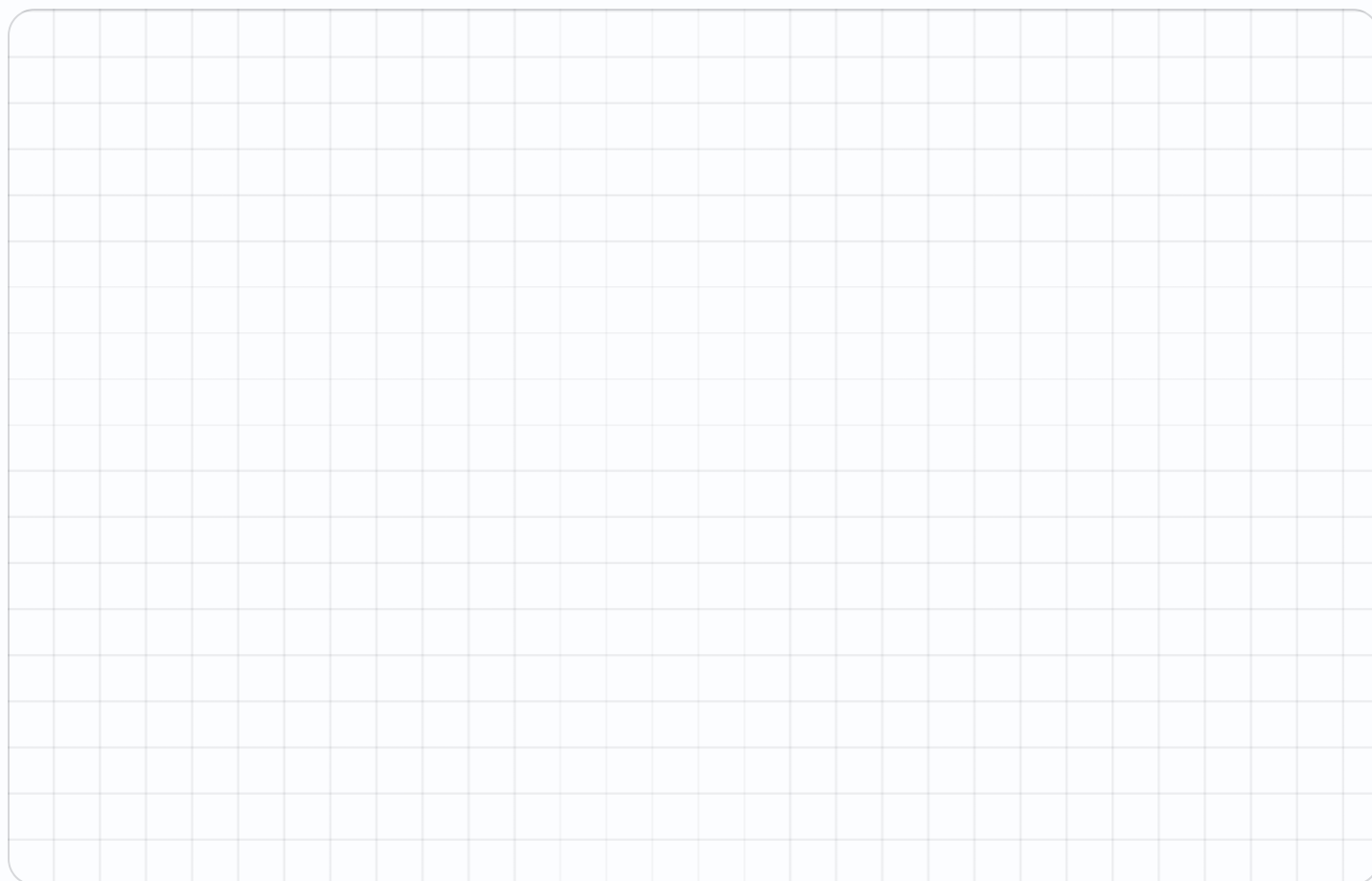
23. Square with corners (0,0), (5,0), (5,5) — the fourth corner's up number (It sits at (0, 5)) (It sits at (0, 5)) (It sits at (0, 5))

24. Distance from (1,2) to (1,9) plus (3,4) to (8,4) (It sits at (0, 5)) (It sits at (0, 5)) (7 + 5)

25. Among rectangles with perimeter 32, the biggest area (It sits at (0, 5)) (It sits at (0, 5))

26. With 100 m of fence, the biggest whole-number area (It sits at (0, 5)) (It sits at (0, 5))

Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE



! Error journal ONE SENTENCE PER MISTAKE

Sorting Shapes & Symmetry

Sort by what is true about a shape, not by what it looks like.

Warm-Up 12 ITEMS • RECALL

1 Sides on a hexagon

2 Sides on a pentagon

3 Right angles in a rectangle

4 Equal sides on an equilateral triangle

5 Sides on an octagon

6 Pairs of parallel sides in a rectangle

7 A 4 by 2 plus a 4 by 2 — total area

8 A 5 by 5 plus a 3 by 3

9 A 6 by 3 plus a 6 by 1

10 A 7 by 2 plus a 4 by 2

11 A 12 by 1 plus a 6 by 2

12 A 8 by 3 plus a 2 by 3

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. Lines of symmetry in a square (Two through the middle, two through the corners)

.....

14. Lines of symmetry in a rectangle (not a square) (The corner folds don't match)

.....

15. Lines of symmetry in an equilateral triangle (One from each corner)

.....

16. Lines of symmetry in a circle — type 0 if too many to count (Infinitely many, so type 0)

.....

17. A 9 by 7 with a 3 by 2 removed — area

.....

18. A 12 by 12 with a 5 by 5 removed

.....

★ Challenge NOT GRADED • REASON IT OUT

22. Degrees in a quarter turn (A right angle)

.....

23. Lines of symmetry in a regular hexagon (One per pair of opposite corners and sides)

.....

24. Sides of a shape whose angles add to 360 (Any quadrilateral)

.....

25. A 25 by 18 room with a 5 by 6 alcove added — total area

.....

26. A 15 by 15 with a 15 by 4 strip removed

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal ONE SENTENCE PER MISTAKE

Angle Guess & Map a Planet

Enrichment day. Estimate before you measure, every single time.

✦ Warm-Up 2 ITEMS • RECALL

1 With 20 m and no wall,
the biggest area

2 Sides you must fence
with a wall behind you

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

3. 20 m against a wall, 5 m deep each side — the width

.....

4. That pen's area

.....

5. Angle on a straight line next to 47° (180 - 47)
6. Angles round a point: 100° , 130° and $?^\circ$ (360 - 230)
7. Rectangle perimeter 20 with the largest area — that area (The 5 by 5 square)
8. Rectangle area 36 with the smallest perimeter — that perimeter (6 by 6)
9. Distance from (3,7) to (11,7) (11 - 3)
10. A shape with 4 lines of symmetry and 4 equal sides — its angle count (It's a square)
11. Turn 45° eight times — total degrees (45×8)
12. Interior angles of a triangle, added (Always)
13. 20 m against a wall, the biggest area
14. 32 m against a wall, the biggest area
15. How much bigger is that than the no-wall best for 32 m

Week 1 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

1.1 • What an Angle Is

OUT OF 13

Warm-Up • 1-8 • 4 | 2 | 14 | 16 | 22 | 28 | 20 | 8

Core • 9-13 • 40 | 6 | 5 | 62 | 16

Challenge • 14-25 • 90 | 180 | 360 | 45 | 50 | 150 | 50 | 20 | 90 | 360 | 360 | 6

1.2 • Perimeter & Area

OUT OF 18

Warm-Up • 1-12 • 16 | 15 | 24 | 36 | 24 | 20 | 10 | 16 | 24 | 49 | 9 | 30

Core • 13-18 • 8 | 6 | 84 | 38 | 30 | 84

Challenge • 23-26 • 36 | 11 | 43 | 72

1.3 • The Coordinate Plane

OUT OF 18

Warm-Up • 1-12 • 3 | 5 | 7 | 0 | 6 | 1 | 20 | 9 | 20 | 25 | 16 | 21

Core • 13-18 • 5 | 7 | 16 | 15 | 5 | 25

Challenge • 23-26 • 5 | 12 | 64 | 625

1.4 • Sorting Shapes & Symmetry

OUT OF 18

Warm-Up • 1-12 • 6 | 5 | 4 | 3 | 8 | 2 | 16 | 34 | 24 | 22 | 24 | 30

Core • 13-18 • 4 | 2 | 3 | 0 | 57 | 119

Challenge • 22-26 • 90 | 6 | 4 | 480 | 165

Fri • Angle Guess & Map a Planet

OUT OF 4

Warm-Up • 1-2 • 25 | 3

Core • 3-4 • 10 | 50

Challenge • 5-15 • 133 | 130 | 25 | 24 | 8 | 4 | 360 | 180 | 50 | 128 | 64

Perimeter

The distance all the way round. Add every side, or be clever about it.

Warm-Up 12 ITEMS • RECALL

1 Perimeter of a 4 by 3 rectangle

2 Perimeter of a 5 by 5 square

3 Perimeter of a 10 by 2 rectangle

4 Perimeter of a 6 by 6 square

5 Perimeter of a 7 by 1 rectangle

6 Perimeter of a 3 by 3 square

7 Perimeter of a 6 by 2 rectangle

8 Perimeter of a 8 by 8 square

9 Perimeter of a 12 by 1 rectangle

10 Perimeter of a 9 by 9 square

11 Perimeter of a 5 by 4 rectangle

12 Perimeter of a 10 by 10 square

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Perimeter of a 12 by 8 rectangle
.....

14. A square of perimeter 36 — one side
.....

15. A rectangle of perimeter 20, one side 6 — the other
.....

16. Perimeter of a 15 by 9 rectangle
.....

17. A square of perimeter 100 — one side
.....

18. Perimeter of a 16 by 9 rectangle
.....

★ Challenge NOT GRADED • REASON IT OUT

23. A rectangle of perimeter 30 with whole sides — how many different ones

.....

24. Perimeter of an L made from a 5 by 5 square with a 2 by 2 corner removed

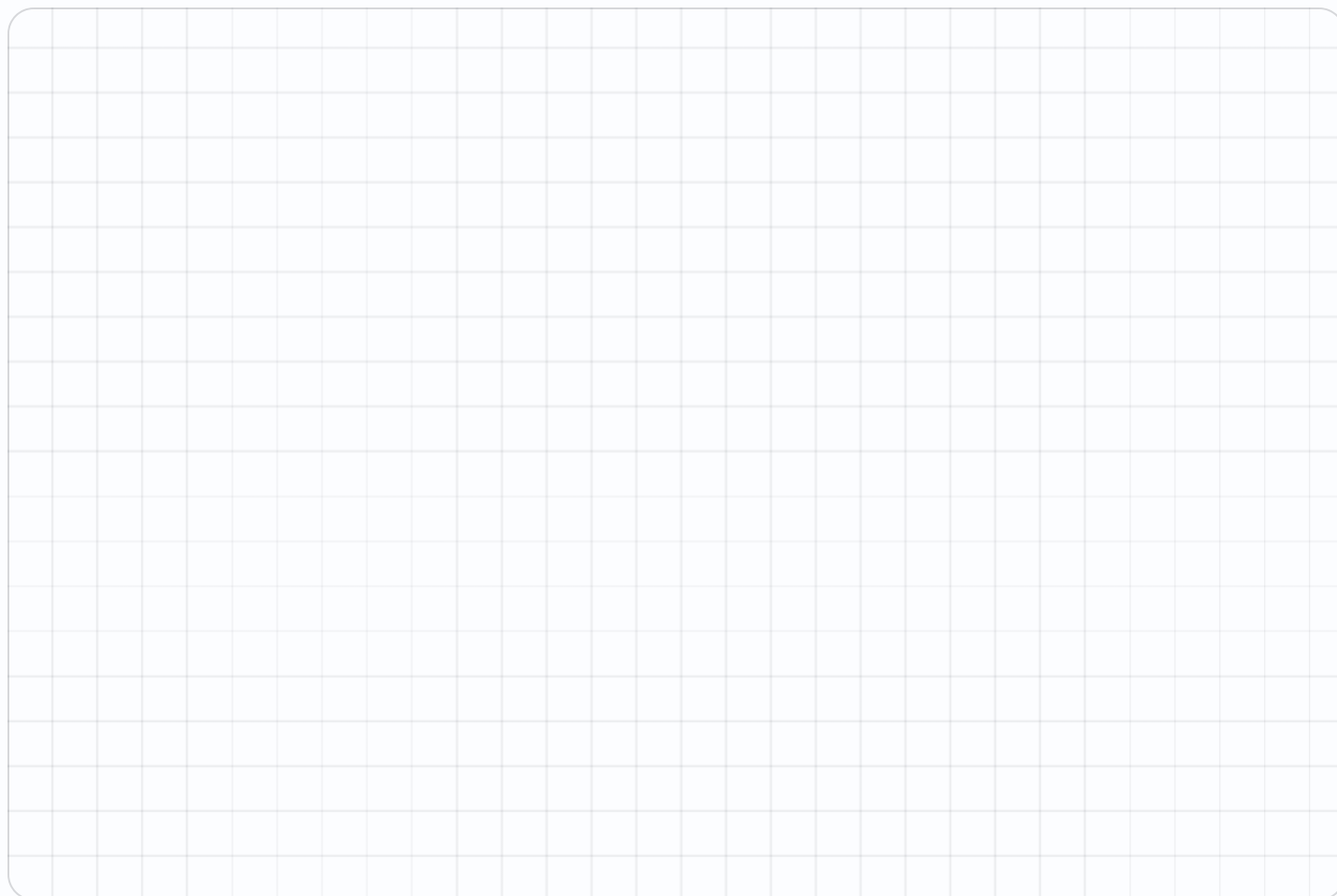
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25. A rectangle of perimeter 40 with whole sides — how many different ones

.....

26. Perimeter of an L from a 8 by 8 square with a 3 by 3 corner removed

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Area

Length times width — and count the squares if you doubt it.

Warm-Up 12 ITEMS • RECALL

1 Area of a 4 by 3 rectangle

2 Area of a 5 by 5 square

3 Area of a 10 by 2 rectangle

4 Area of a 6 by 6 square

5 Area of a 7 by 1 rectangle

6 Area of a 8 by 3 rectangle

7 Area of a 6 by 2 rectangle

8 Area of a 8 by 8 square

9 Area of a 12 by 1 rectangle

10 Area of a 9 by 9 square

11 Area of a 5 by 4 rectangle

12 Area of a 10 by 7 rectangle

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Area of a 12 by 8 rectangle

.....

14. Area 48, one side 6 — the other

.....

15. Area of a 15 by 9 rectangle

.....

16. A square of area 49 — one side

.....

17. Area 144 with equal sides — one side

.....

18. Area of a 16 by 9 rectangle

.....

★ Challenge NOT GRADED • REASON IT OUT

23. Area of an L: a 6 by 6 square minus a 2 by 3 corner

.....

24. Double both sides of a 4 by 5 rectangle — the new area

.....

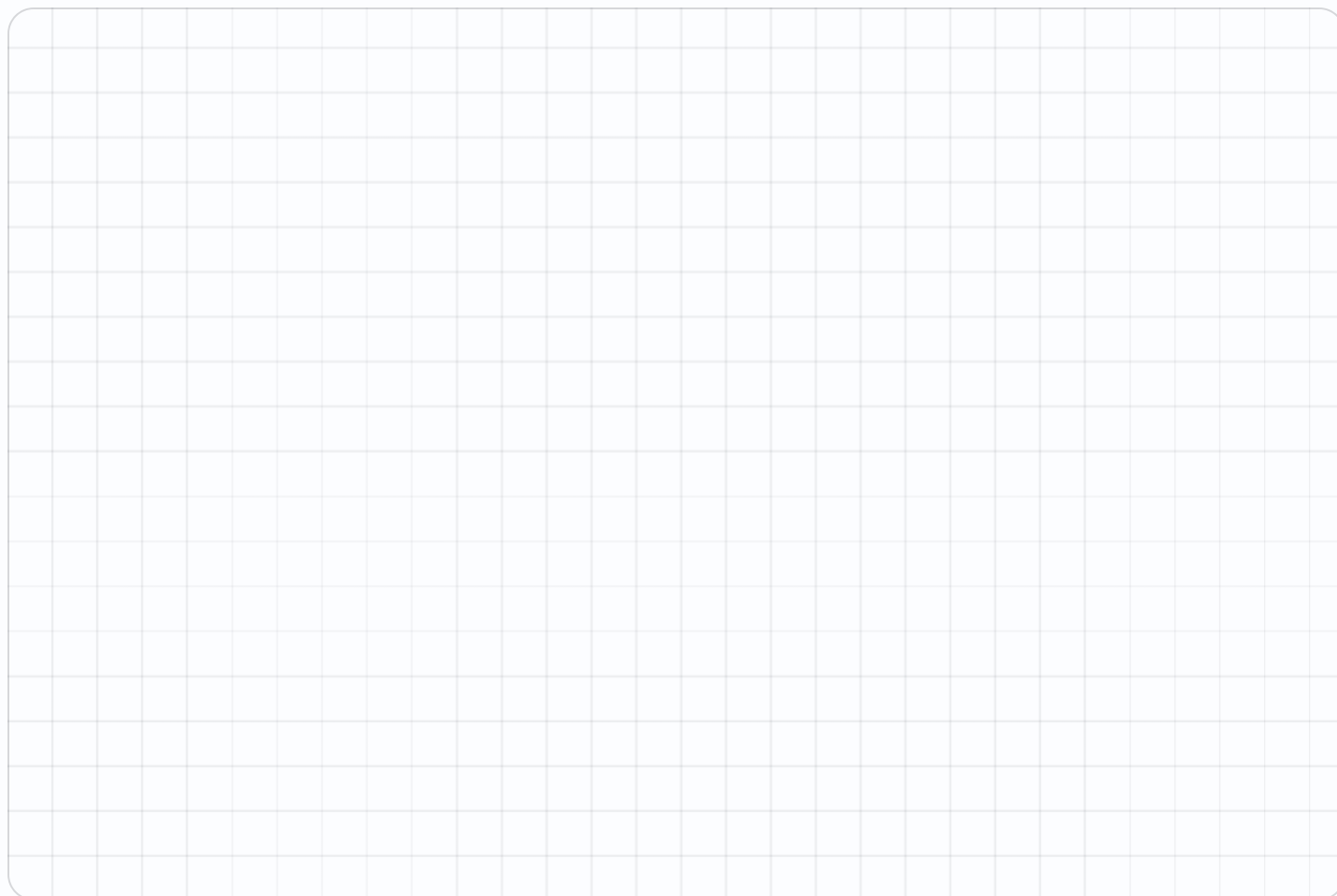
25. Area of an L: a 8 by 8 square minus a 3 by 4 corner

.....

26. Double both sides of a 5 by 7 rectangle — the new area

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE



! Error journal ONE SENTENCE PER MISTAKE

Same Fence, Different Field

Perimeter fixed, area wildly different. That is the Big Question.

Warm-Up 12 ITEMS • RECALL

1 A 1 by 11 rectangle —
perimeter

2 That rectangle — area

3 A 6 by 6 square —
perimeter

4 That square — area

5 A 2 by 10 rectangle —
area

6 A 4 by 8 rectangle — area

7 A rectangle cut into 2
equal parts — one part is 1
over what

8 A rectangle cut into 3
equal parts — one part is 1
over what

9 A rectangle cut into 4
equal parts — one part is 1
over what

10 A rectangle cut into 6
equal parts — one part is 1
over what

11 A rectangle cut into 8
equal parts — one part is 1
over what

12 A 1 by 15 rectangle —
perimeter

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

16. With 24 m of fence, the biggest whole-number area

.....

17. With 20 m of fence, the biggest area

.....

18. With 16 m of fence, the biggest area

.....

19. With 24 m of fence, the smallest area using whole sides

.....

20. With 30 m of fence, the biggest whole-number area

.....

21. A shape of 8 squares split into 2 equal parts — squares in each part

.....

★ Challenge NOT GRADED • REASON IT OUT

32. The shape that always gives the biggest area for a fixed perimeter — type square or circle

33. Among rectangles with perimeter 40, the biggest area

34. A 12-square shape split into 3 equal parts, then one part split in 2 — squares in the smallest piece

35. The shape giving the biggest area for a fixed perimeter — type square or circle

36. Among rectangles with perimeter 60, the biggest area

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Area of Compound Shapes

Split it into rectangles, do each, add them up.

Warm-Up 12 ITEMS • RECALL

1 A 3 by 2 plus a 3 by 2 —
total area

2 A 4 by 4 plus a 2 by 2

3 A 5 by 3 plus a 5 by 1

4 A 6 by 2 plus a 3 by 2

5 A 10 by 1 plus a 5 by 2

6 A 7 by 3 plus a 1 by 3

7 A 4 by 3 plus a 4 by 3 —
total area

8 A 6 by 6 plus a 3 by 3

9 A 7 by 4 plus a 7 by 1

10 A 9 by 2 plus a 4 by 2

11 A 15 by 1 plus a 5 by 3

12 A 6 by 4 plus a 2 by 4

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. An 8 by 6 with a 2 by 3 removed — area

.....

14. A 10 by 10 with a 4 by 4 removed

.....

15. An L of a 9 by 4 and a 5 by 3

.....

16. A 12 by 5 with a 3 by 5 removed

.....

17. A T of a 8 by 2 and a 2 by 6

.....

18. A 10 by 8 with a 3 by 4 removed — area

.....

★ Challenge NOT GRADED • REASON IT OUT

23. A 20 by 15 room with a 4 by 5 alcove added — total area

.....

24. A 12 by 12 with a 12 by 3 strip removed

.....

25. A 30 by 20 room with a 6 by 5 alcove added — total area

.....

26. A 20 by 20 with a 20 by 6 strip removed

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

The Biggest Pen

24 metres of fence, and a wall to build against. Now what?

NAME _____

DATE _____

RIGHT ____ OF ____ TRIED

✦ Warm-Up

4 ITEMS • RECALL

- 1 With 24 m and no wall, the biggest area

- 2 Sides you must fence with a wall behind you

- 3 With 36 m and no wall, the biggest area

- 4 Fenced sides with a wall behind

◆ Core

GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. 24 m against a wall, 8 m deep each side — the width

.....
6. That pen's area

.....
7. 36 m against a wall, 9 m deep each side — the width

.....

Mark the reasoning, not the handwriting.

Week 2 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

2.1 • Perimeter

OUT OF 18

Warm-Up • 1-12 • 14 | 20 | 24 | 24 | 16 | 12 | 16 | 32 | 26 | 36 | 18 | 40

Core • 13-18 • 40 | 9 | 4 | 48 | 25 | 50

Challenge • 23-26 • 7 | 20 | 10 | 32

2.2 • Area

OUT OF 18

Warm-Up • 1-12 • 12 | 25 | 20 | 36 | 7 | 24 | 12 | 64 | 12 | 81 | 20 | 70

Core • 13-18 • 96 | 8 | 135 | 7 | 12 | 144

Challenge • 23-26 • 30 | 80 | 52 | 140

2.3 • Same Fence, Different Field

OUT OF 18

Warm-Up • 1-12 • 24 | 11 | 24 | 36 | 20 | 32 | 2 | 3 | 4 | 6 | 8 | 32

Core • 16-21 • 36 | 25 | 16 | 11 | 56 | 4

Challenge • 32-36 • circle | 100 | 2 | circle | 225

2.4 • Area of Compound Shapes

OUT OF 18

Warm-Up • 1-12 • 12 | 20 | 20 | 18 | 20 | 24 | 24 | 45 | 35 | 26 | 30 | 32

Core • 13-18 • 42 | 84 | 51 | 45 | 28 | 68

Challenge • 23-26 • 320 | 108 | 630 | 280

Fri • The Biggest Pen

OUT OF 7

Warm-Up • 1-4 • 36 | 3 | 81 | 3

Core • 5-7 • 8 | 64 | 18

Challenge • 8-13 • 72 | 72 | 36 | 162 | 200 | 100

Plotting Points

Across the hall, then up the stairs. Type coordinates like 3,5.

Warm-Up 12 ITEMS • RECALL

1 In (3, 5), the number across

2 In (3, 5), the number up

3 The origin — type as a,b

4 4 across, 2 up — type as a,b

5 0 across, 6 up — type as a,b

6 In (7, 1), the first number

7 In (5, 2), the number across

8 In (5, 2), the number up

9 6 across, 0 up — type as a,b

10 2 across, 7 up — type as a,b

11 0 across, 9 up — type as a,b

12 In (4, 8), the first number

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 3 right of (2,5) — type as a,b

.....

14. 2 up from (4,1) — type as a,b

.....

15. Distance from (2,3) to (7,3)

.....

16. Distance from (4,1) to (4,8)

.....

17. Halfway between (0,0) and (8,0) — type as a,b

.....

18. 5 right of (3,4) — type as a,b

.....

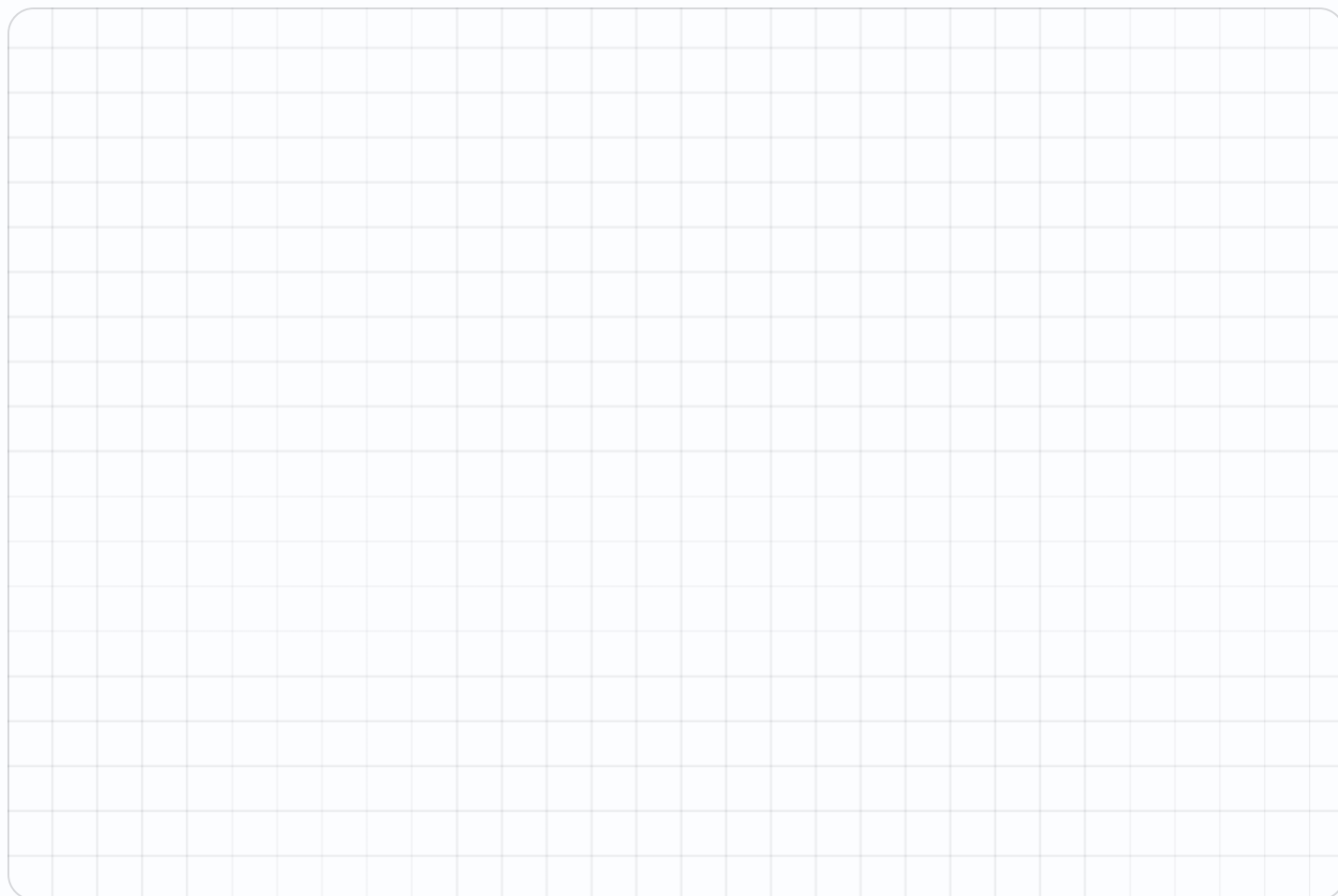
★ Challenge NOT GRADED • REASON IT OUT

23. Halfway between (2,2) and (8,6) — type as a,b

24. From (0,0) to (6,8) going only across and up — total steps

25. Halfway between (1,3) to (9,7) — type as a,b

26. From (0,0) to (7,9) going only across and up — total steps

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Shapes on the Grid

Plot the corners, then read off what you built.

Warm-Up 12 ITEMS • RECALL

1 Corners on a rectangle

2 Corners on a triangle

3 (1,1) to (5,1) — the length

4 (1,1) to (1,4) — the length

5 Sides on a square

6 (0,0) to (3,0) — the length

7 Corners on a square

8 Corners on a pentagon

9 (2,1) to (7,1) — the length

10 (2,1) to (2,6) — the length

11 Sides on a rectangle

12 (0,0) to (5,0) — the length

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. (1,1), (1,5), (6,5), (6,1) — the area

.....

14. That shape's perimeter

.....

15. (2,2), (2,7), (5,7), (5,2) — the area

.....

16. (0,0), (0,4), (4,4), (4,0) — the shape's name

.....

17. The missing corner of a rectangle at (1,1),(1,4),(6,4) — type as a,b

.....

18. (1,1), (1,6), (5,6), (5,1) — the area

.....

★ Challenge NOT GRADED • REASON IT OUT

22. $(0,0)$, $(8,0)$, $(8,5)$, $(0,5)$ — the area

.....

23. A square with corners $(2,2)$ and $(7,7)$ opposite — its area

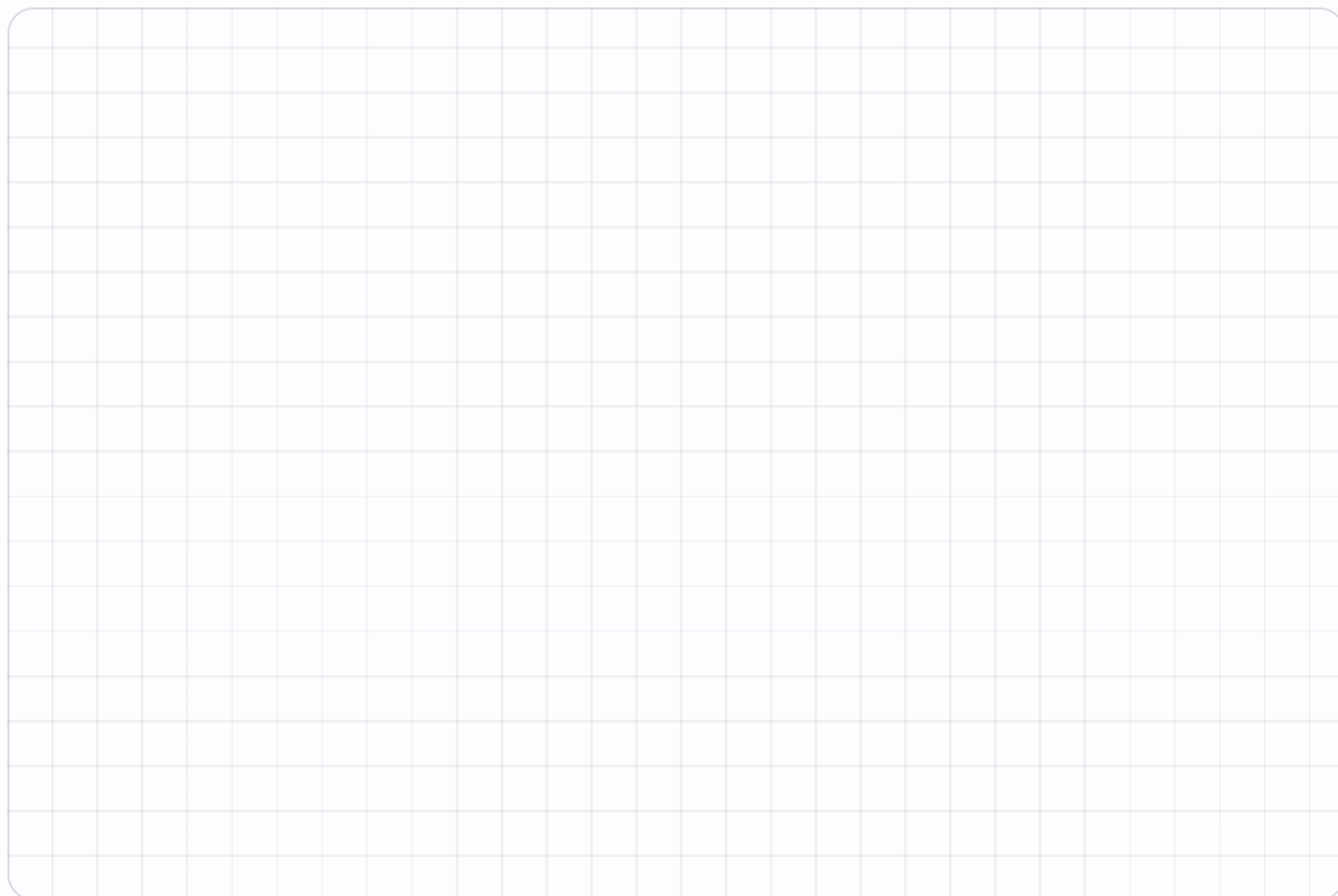
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24. $(0,0)$, $(10,0)$, $(10,6)$, $(0,6)$ — the area

.....

25. A square with corners $(1,1)$ and $(8,8)$ opposite — its area

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Reading a Map

Coordinates are directions somebody else has to follow.

Warm-Up 12 ITEMS • RECALL

1 From (0,0), move 3 across — type as a,b

2 From (3,0), move 2 up — type as a,b

3 From (5,5), move 1 across — type as a,b

4 From (2,2), move 2 up — type as a,b

5 Blocks from (0,0) to (4,0)

6 Blocks from (0,0) to (0,7)

7 From (0,0), move 4 across — type as a,b

8 From (4,0), move 3 up — type as a,b

9 From (6,6), move 2 across — type as a,b

10 From (3,3), move 4 up — type as a,b

11 Blocks from (0,0) to (6,0)

12 Blocks from (0,0) to (0,9)

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. From (1,2) to (6,2) then up 3 — type the end as a,b

14. Total blocks travelled on that route

15. A park at (3,4), a school at (9,4) — blocks apart

16. From (2,1) to (2,9) — blocks

17. From (0,0) to (5,5) across-and-up — blocks

18. From (2,3) to (8,3) then up 4 — type the end as a,b

★ Challenge NOT GRADED • REASON IT OUT

22. A route (1,1) to (7,1) to (7,6) — total blocks

.....

23. The shortest across-and-up route from (2,3) to (9,8) — blocks

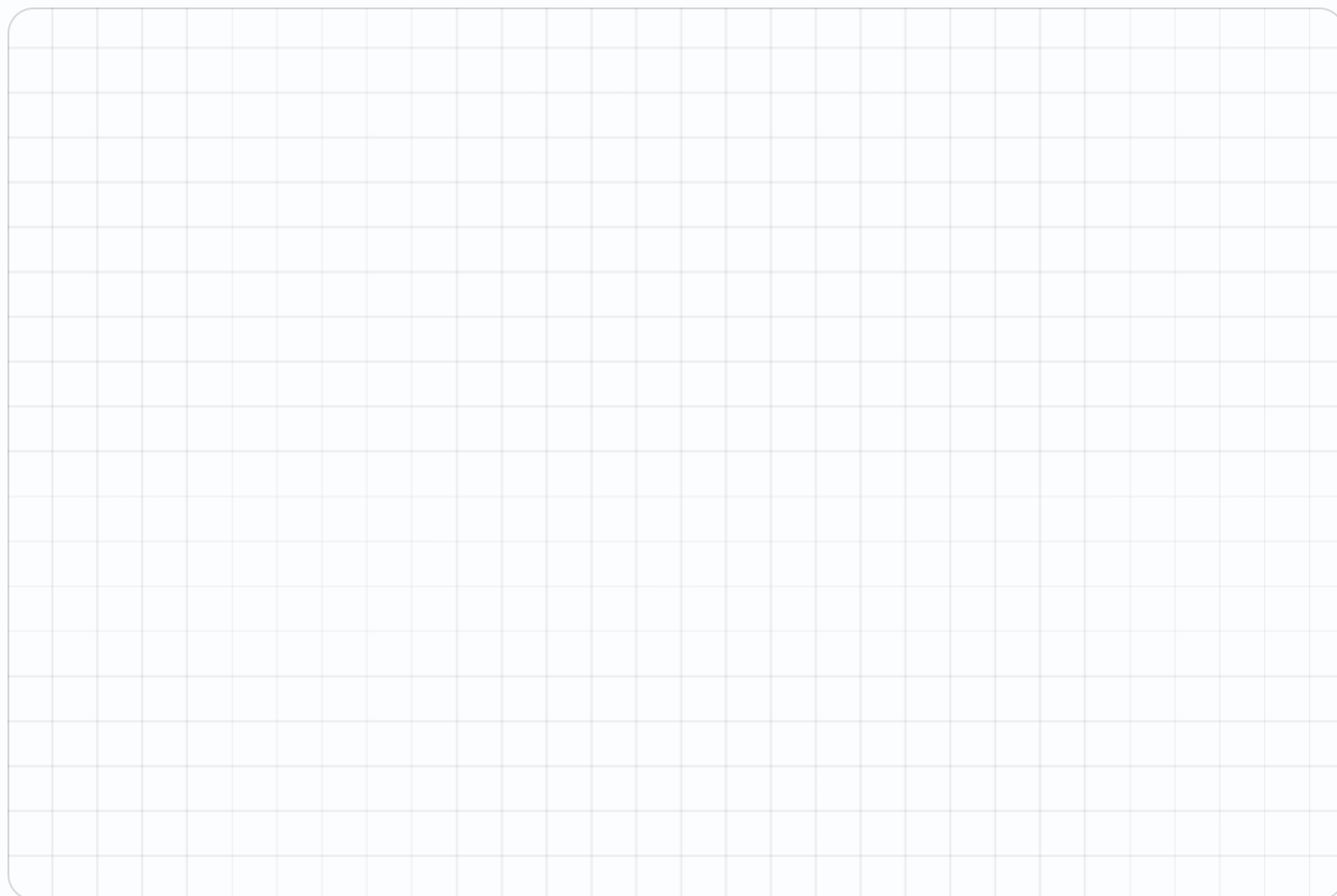
.....

24. A route (2,2) to (9,2) to (9,8) — total blocks

.....

25. The shortest across-and-up route from (1,2) to (10,9) — blocks

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Order Matters

(3,5) and (5,3) are different places. Prove it every time.

Warm-Up 12 ITEMS • RECALL

1 In (3,5), the across value

2 In (5,3), the across value

3 Are (3,5) and (5,3) the same — yes or no

4 In (0,4), the across value

5 In (4,0), the up value

6 A point on the bottom line has which value 0 — type across or up

7 In (4,7), the across value

8 In (7,4), the across value

9 Are (4,7) and (7,4) the same — yes or no

10 In (0,5), the across value

11 In (5,0), the up value

12 A point on the left edge has which value 0 — type across or up

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. (2,6) and (6,2) — how far apart across

.....

14. Those two points — how far apart up

.....

15. A point with the same across and up as (5,5) — type as a,b

.....

16. Points where across equals up, from (0,0) to (5,5) — how many

.....

17. Swap (7,2) — type the new point as a,b

.....

18. (3,8) and (8,3) — how far apart across

.....

★

Challenge

NOT GRADED · REASON IT OUT

22. Points on the line where across equals up, from 0 to 10 — how many

.....

23. (1,4), (4,1), (1,1) — the area of that triangle

.....

24. Points on the line where across equals up, from 0 to 20 — how many

.....

25. (2,5), (5,2), (2,2) — the area of that triangle

.....

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

!

Error journal

ONE SENTENCE PER MISTAKE

Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

Warm-Up 4 ITEMS • RECALL

1 Perimeter of a 4 by 3 rectangle

2 Area of a 4 by 3 rectangle

3 Perimeter of a 5 by 4 rectangle

4 Area of a 5 by 4 rectangle

Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. Area 48, one side 6 — the other
6. With 20 m of fence, the biggest area
7. An 8 by 6 with a 2 by 3 removed — area
8. Distance from (2,3) to (7,3)
9. (1,1), (1,5), (6,5), (6,1) — the area
10. Area 72, one side 8 — the other
11. With 40 m of fence, the biggest area
12. A 10 by 8 with a 3 by 4 removed — area
13. Distance from (1,5) to (9,5)
14. (1,1), (1,6), (5,6), (5,1) — the area

★ Challenge NOT GRADED • REASON IT OUT

15. A rectangle of perimeter 30 with whole sides — how many different ones

.....

16. A rectangle of perimeter 40 with whole sides — how many different ones

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Week 3 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

3.1 • Plotting Points

OUT OF 18

Warm-Up • 1-12 • 3 | 5 | 0,0 | 4,2 | 0,6 | 7 | 5 | 2 | 6,0 | 2,7 | 0,9 | 4

Core • 13-18 • 5,5 | 4,3 | 5 | 7 | 4,0 | 8,4

Challenge • 23-26 • 5,4 | 14 | 5,5 | 16

3.2 • Shapes on the Grid

OUT OF 18

Warm-Up • 1-12 • 4 | 3 | 4 | 3 | 4 | 3 | 4 | 5 | 5 | 5 | 4 | 5

Core • 13-18 • 20 | 18 | 15 | square | 6,1 | 20

Challenge • 22-25 • 40 | 25 | 60 | 49

3.3 • Reading a Map

OUT OF 18

Warm-Up • 1-12 • 3,0 | 3,2 | 6,5 | 2,4 | 4 | 7 | 4,0 | 4,3 | 8,6 | 3,7 | 6 | 9

Core • 13-18 • 6,5 | 8 | 6 | 8 | 10 | 8,7

Challenge • 22-25 • 11 | 12 | 13 | 16

3.4 • Order Matters

OUT OF 18

Warm-Up • 1-12 • 3 | 5 | no | 0 | 0 | up | 4 | 7 | no | 0 | 0 | across

Core • 13-18 • 4 | 4 | 5,5 | 6 | 2,7 | 5

Challenge • 22-25 • 11 | 4.5 | 21 | 4.5

Fri • Mid-Unit Quiz

OUT OF 14

Warm-Up • 1-4 • 14 | 12 | 18 | 20

Core • 5-14 • 8 | 25 | 42 | 5 | 20 | 9 | 100 | 68 | 8 | 20

Challenge • 15-16 • 7 | 10

Lines of Symmetry

Fold it. If both halves land exactly, that fold is a line of symmetry.

Warm-Up 11 ITEMS • RECALL

1 Lines of symmetry in a square

2 In a rectangle that is not a square

3 In a circle — type the word

4 In an equilateral triangle

5 In the letter A

6 In the letter H

7 Lines of symmetry in a rectangle

8 In the letter O — type the word

9 In the letter T

10 In the letter X

11 In a square

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. In a regular pentagon
.....

13. In a regular hexagon
.....

14. In an isosceles triangle
.....

15. In a scalene triangle
.....

16. In a rhombus that is not a square
.....

17. In a regular heptagon
.....

★ Challenge NOT GRADED • REASON IT OUT

22. In a regular octagon

.....

23. In a regular polygon with 12 sides

.....

24. In a regular polygon with 20 sides

.....

25. In a semicircle

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Flips and Turns

Slide, flip, turn. The shape keeps its size and its angles.

✦ Warm-Up 4 ITEMS • RECALL

1 Quarter turns in a full turn

2 Half turns in a full turn

3 Quarter turns in a half turn

4 Half turns in two full turns

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

5. (2,3) flipped over the vertical line through 0 — the across value becomes

.....

6. Slide (2,3) four right — type as a,b

.....

7. (3,4) flipped over the vertical line through 0 — the across value becomes

.....

8. Slide (3,4) five up — type as a,b

.....

20. An equilateral triangle turned 120° looks the same — yes or no

Angles

Right, acute, obtuse — and what they add to.

Warm-Up 12 ITEMS • RECALL

1 Angles in a triangle add to _____

2 A 4 by 3 rectangle joined to a 4 by 2 — total squares _____

3 A 5 by 2 rectangle joined to a 5 by 4 — total squares _____

4 A 6 by 3 rectangle joined to a 6 by 3 — total squares _____

5 A 3 by 5 rectangle joined to a 3 by 5 — total squares _____

6 A 7 by 2 rectangle joined to a 7 by 3 — total squares _____

7 A rectangle 3 squares across and 4 down — squares altogether _____

8 A rectangle 5 squares across and 2 down — squares altogether _____

9 A rectangle 6 squares across and 3 down — squares altogether _____

10 A rectangle 4 squares across and 4 down — squares altogether _____

11 A rectangle 7 squares across and 2 down — squares altogether _____

12 A rectangle 5 squares across and 5 down — squares altogether _____

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. Angles in a quadrilateral add to

14. $4 \times (3 + 2)$ — work out the bracket first

15. 4×3 plus 4×2

16. $5 \times (2 + 4)$ — work out the bracket first

17. 5×2 plus 5×4

18. $6 \times (3 + 3)$ — work out the bracket first

Classifying Shapes

Sides, angles, parallels. Name it from its properties.

Warm-Up 12 ITEMS • RECALL

1 Sides on a pentagon

2 Sides on a hexagon

3 Sides on an octagon

4 Sides on a quadrilateral

5 Equal sides on an equilateral triangle

6 Sides on a triangle

7 Sides on a heptagon

8 Sides on a nonagon

9 Sides on a decagon

10 Sides on a rhombus

11 Equal sides on an isosceles triangle

12 Sides on a square

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Is every square a rectangle — yes or no

.....

14. Is every rectangle a square

.....

15. Pairs of parallel sides in a parallelogram

.....

16. Pairs of parallel sides in a trapezoid

.....

17. Is every rhombus a parallelogram — yes or no

.....

18. Is every parallelogram a rhombus

.....



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Tessellation

Which shapes tile a floor with no gaps? Test three and explain.

✦ Warm-Up 4 ITEMS • RECALL

1 Do squares tile with no gaps — yes or no

2 Do circles tile with no gaps

3 Do rectangles tile with no gaps — yes or no

4 Do regular octagons tile alone

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. Do equilateral triangles tile — yes or no

.....

6. Do regular hexagons tile

.....

7. Do parallelograms tile — yes or no

.....

8. Do regular heptagons tile

.....



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Week 4 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

4.1 • Lines of Symmetry

OUT OF 17

Warm-Up • 1-11 • 4 | 2 | infinite | 3 | 1 | 2 | 2 | infinite | 1 | 2 | 4

Core • 12-17 • 5 | 6 | 1 | 0 | 2 | 7

Challenge • 22-25 • 8 | 12 | 20 | 1

4.2 • Flips and Turns

OUT OF 8

Warm-Up • 1-4 • 4 | 2 | 2 | 4

Core • 5-8 • -2 | 6,3 | -3 | 3,9

Challenge • 9-20 • 90 | 180 | 360 | 270 | yes | 4 | 180 | 120 | 60 | 180 | 270 |
yes

4.3 • Angles

OUT OF 18

Warm-Up • 1-12 • 180 | 20 | 30 | 36 | 30 | 35 | 12 | 10 | 18 | 16 | 14 | 25

Core • 13-18 • 360 | 20 | 20 | 30 | 30 | 36

Challenge • 43-54 • 90 | 180 | 360 | acute | obtuse | 120 | 60 | 90 | 160 | 60 |
120 | 64

4.4 • Classifying Shapes

OUT OF 18

Warm-Up • 1-12 • 5 | 6 | 8 | 4 | 3 | 3 | 7 | 9 | 10 | 4 | 2 | 4

Core • 13-18 • yes | no | 2 | 1 | yes | no

Challenge • 21-26 • right | square | scalene | acute | rhombus | isosceles

Fri • Tessellation

OUT OF 8

Warm-Up • 1-4 • yes | no | yes | no

Core • 5-8 • yes | yes | yes | no

Challenge • 9-14 • no | 360 | 3 | 4 | 6 | 36

Map a Planet

An invented world on the coordinate plane, with real distances.

Warm-Up 11 ITEMS • RECALL

1 A harbour at (2,3) — the across value

2 A peak at (7,7) — the up value

3 Distance from (2,3) to (6,3)

4 Distance from (5,1) to (5,6)

5 The origin — type as a,b

6 3 across, 8 up — type as a,b

7 A dock at (3,4) — the across value

8 A peak at (8,9) — the up value

9 Distance from (3,4) to (8,4)

10 Distance from (6,2) to (6,9)

11 5 across, 5 up — type as a,b

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. Sailing (1,1) to (9,1) then to (9,5) — total distance

13. A square island (2,2) to (8,8) — its area

14. That island's perimeter

15. Halfway between (0,0) and (10,4) — type as a,b

16. A route (0,0) to (4,0) to (4,7) — distance

17. Sailing (2,2) to (10,2) then to (10,7) — total distance

★

Challenge

NOT GRADED · REASON IT OUT

21. A rectangular sea from (1,1) to (11,6) — its area
-
22. Ten landmarks each 3 units apart in a line — total length
-
23. A rectangular sea from (2,2) to (14,8) — its area
-
24. Twelve landmarks each 4 units apart in a line — total length
-

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

!

Error journal

ONE SENTENCE PER MISTAKE

Sailing Directions

Somebody else has to follow them. Ambiguity is the enemy.

Warm-Up 11 ITEMS • RECALL

1 From (0,0) move 5 across — type as a,b

2 Then 3 up — type as a,b

3 Total distance travelled

4 From (4,4) move 2 up — type as a,b

5 From (4,4) move 2 across — type as a,b

6 Blocks from (0,0) to (3,4) across-and-up

7 From (0,0) move 7 across — type as a,b

8 Then 2 up — type as a,b

9 From (5,5) move 3 up — type as a,b

10 From (5,5) move 3 across — type as a,b

11 Blocks from (0,0) to (5,6) across-and-up

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. (1,1) to (1,7) to (5,7) — total distance

.....

13. (2,2) to (8,2) to (8,9) — total distance

.....

14. A round trip (0,0) to (6,0) and back — distance

.....

15. (3,3) to (3,10) — distance

.....

16. A square patrol of side 5 — total distance

.....

17. (2,1) to (2,9) to (7,9) — total distance

.....

★ Challenge NOT GRADED • REASON IT OUT

22. A patrol (1,1),(7,1),(7,5),(1,5) and back to start — total distance

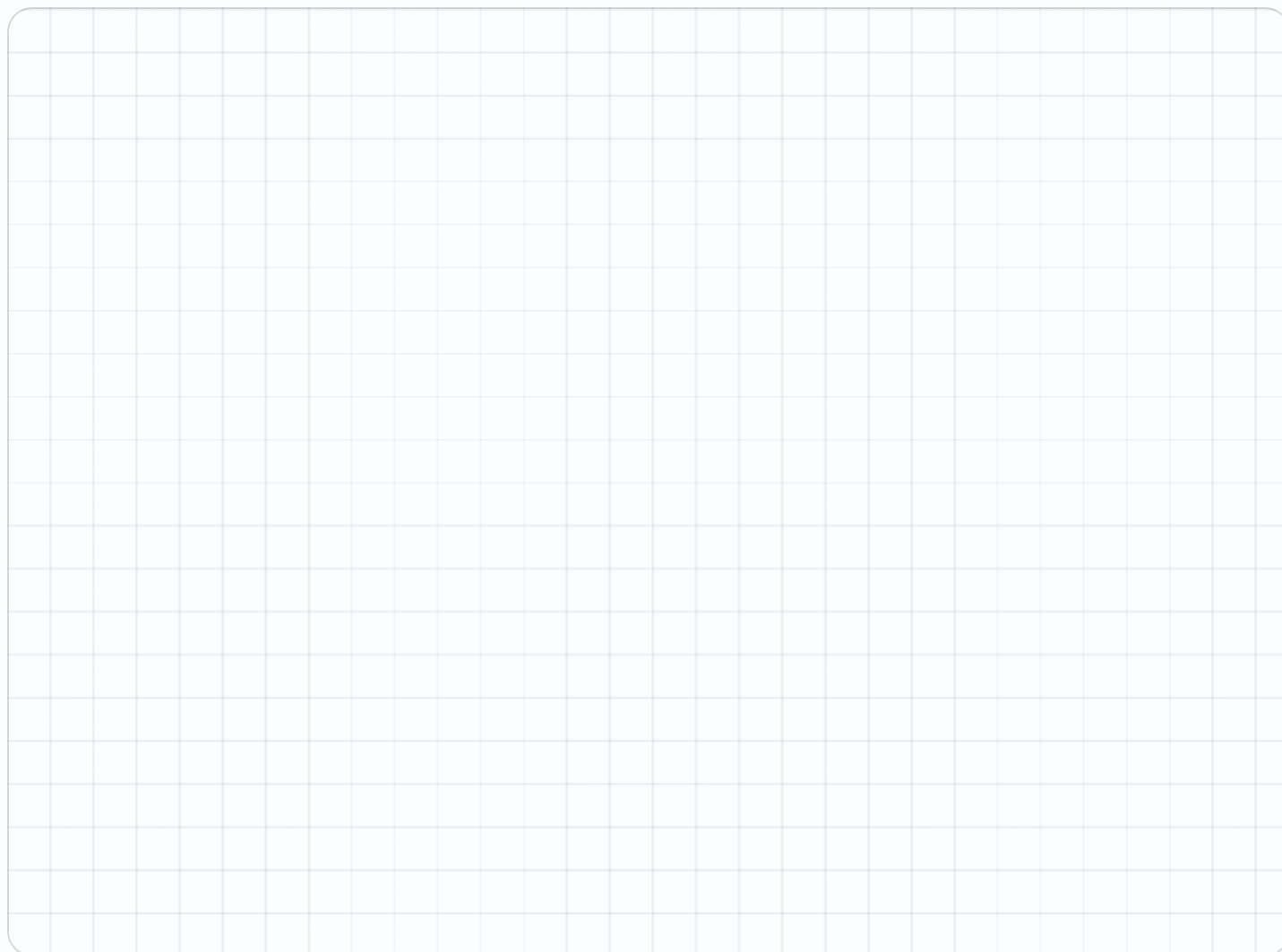
.....

23. The area enclosed by that patrol

.....

24. A patrol (2,2),(9,2),(9,7),(2,7) and back — total distance

.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Mission Review

Everything from five weeks, mixed together.

Warm-Up 10 ITEMS • RECALL

1 Perimeter of a 5 by 5 square

2 Area of a 5 by 5 square

3 Lines of symmetry in a square

4 In (3,5), the up value

5 Angles in a triangle add to

6 Perimeter of a 6 by 6 square

7 Area of a 6 by 6 square

8 Lines of symmetry in an equilateral triangle

9 In (6,2), the up value

10 Angles in a quadrilateral add to

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. Area 96 with one side 12 — the other
.....

12. With 24 m of fence, the biggest area
.....

13. (2,2), (2,7), (5,7), (5,2) — the area
.....

14. Lines of symmetry in a regular hexagon
.....

15. Area 144 with one side 16 — the other
.....

16. With 32 m of fence, the biggest area
.....



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

Error Journal Sweep

Re-read every entry from the mission. Fix only what repeats.

Warm-Up 12 ITEMS • RECALL

1 Perimeter of a 6 by 6 square

2 Area of a 8 by 3 rectangle

3 Sides on a hexagon

4 Distance from (2,3) to (7,3)

5 Lines of symmetry in a rectangle

6 Perimeter of a 8 by 8 square

7 Area of a 9 by 4 rectangle

8 Sides on a decagon

9 Distance from (1,5) to (9,5)

10 Lines of symmetry in a square

11 An 8 by 6 with a 2 by 3 removed — area

12 A square of perimeter 36 — one side

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. (1,1), (1,5), (6,5), (6,1) — the perimeter

.....

14. Is every square a rectangle — yes or no

.....

15. A 10 by 8 with a 3 by 4 removed — area

.....

16. A square of perimeter 48 — one side

.....

17. (1,1), (1,6), (5,6), (5,1) — the perimeter

.....

18. Is every rhombus a parallelogram — yes or no

.....

19. Degrees round a point

.....

20. Two angles on a line, one is 60° — the other

.....

21. Among rectangles with perimeter 40, the biggest area

.....

22. A regular hexagon — smallest turn leaving it unchanged, in degrees

.....

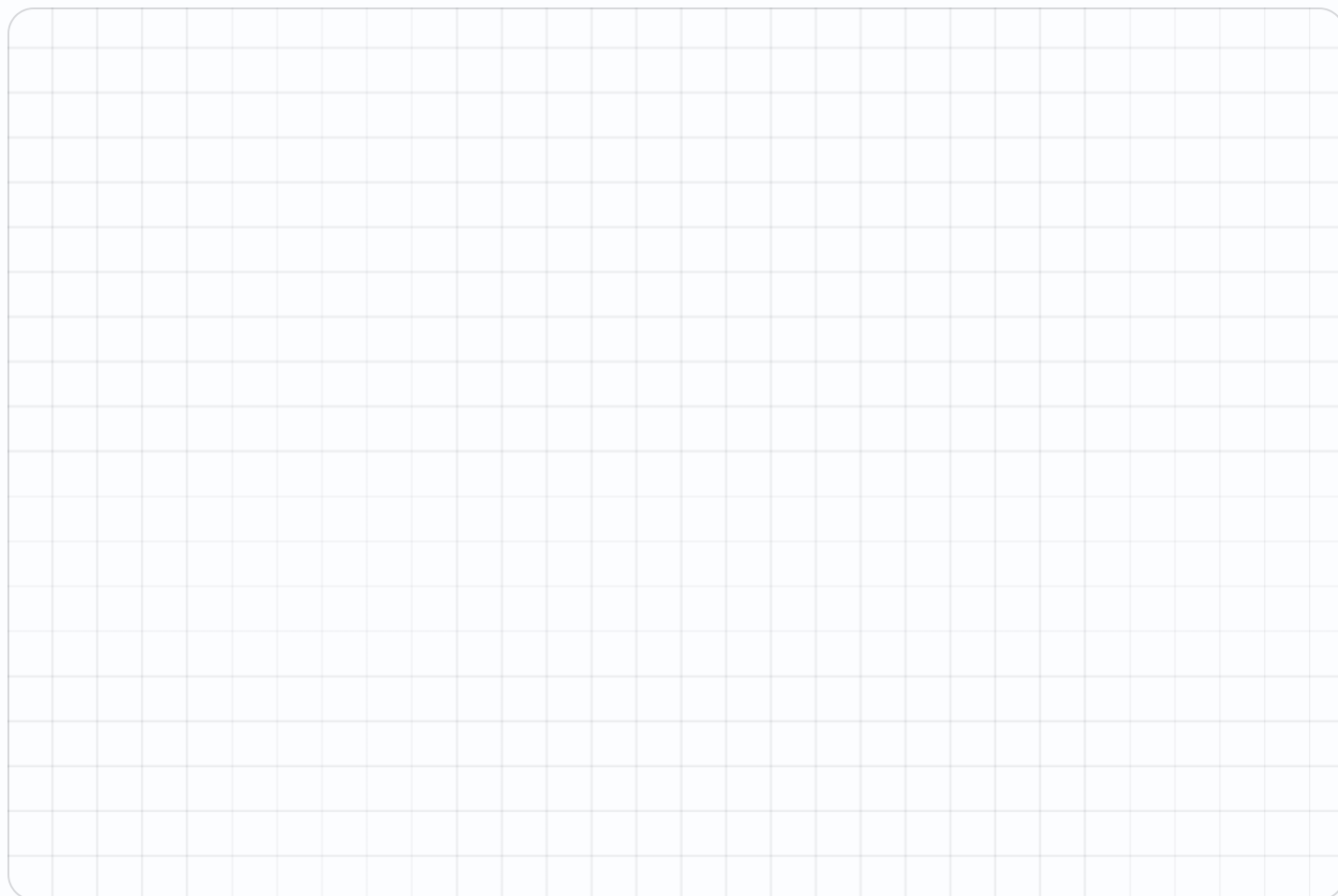
★ Challenge NOT GRADED • REASON IT OUT

23. Degrees in a full turn

24. Two angles on a line, one is 35° — the other

25. Among rectangles with perimeter 60, the biggest area

26. A regular octagon — smallest turn leaving it unchanged, in degrees

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Mission 06 Test

Twelve items plus the Big Question, answered out loud.

NAME _____
DATE _____
RIGHT ____ OF ____ TRIED

Warm-Up 4 ITEMS · RECALL

- 1 Perimeter of a 10 by 2 rectangle

- 2 Area of a 10 by 2 rectangle

- 3 Perimeter of a 12 by 1 rectangle

- 4 Area of a 12 by 1 rectangle

Core GRADED · SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. Area of a 12 by 8 rectangle
.....
6. A rectangle of perimeter 20, one side 6 — the other
.....
7. With 20 m of fence, the biggest area
.....
8. A 12 by 5 with a 3 by 5 removed — area
.....
9. Distance from (4,1) to (4,8)
.....
10. (1,1), (1,5), (6,5), (6,1) — the area
.....
11. Lines of symmetry in a regular pentagon
.....
12. Area of a 16 by 9 rectangle
.....
13. A rectangle of perimeter 26, one side 8 — the other
.....
14. With 40 m of fence, the biggest area
.....
15. A 16 by 7 with a 5 by 7 removed — area
.....
16. Distance from (7,2) to (7,10)
.....
17. (3,2), (3,8), (7,8), (7,2) — the area
.....
18. Lines of symmetry in a regular heptagon
.....
19. A triangle with 90° and 30° — the third angle
.....
20. A 20 by 15 room with a 4 by 5 alcove added — total area
.....

★ Challenge NOT GRADED • REASON IT OUT

21. A patrol (1,1),(7,1),(7,5),(1,5) — the area enclosed

22. A triangle with 90° and 45° — the third angle

23. A 30 by 20 room with a 6 by 5 alcove added — total area

24. A patrol (2,2),(9,2),(9,7),(2,7) — the area enclosed

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Week 5 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

5.1 • Map a Planet

OUT OF 17

Warm-Up • 1-11 • 2 | 7 | 4 | 5 | 0,0 | 3,8 | 3 | 9 | 5 | 7 | 5,5

Core • 12-17 • 12 | 36 | 24 | 5,2 | 11 | 13

Challenge • 21-24 • 50 | 27 | 72 | 44

5.2 • Sailing Directions

OUT OF 17

Warm-Up • 1-11 • 5,0 | 5,3 | 8 | 4,6 | 6,4 | 7 | 7,0 | 7,2 | 5,8 | 8,5 | 11

Core • 12-17 • 10 | 13 | 12 | 7 | 20 | 13

Challenge • 22-24 • 20 | 24 | 24

5.3 • Mission Review

OUT OF 16

Warm-Up • 1-10 • 20 | 25 | 4 | 5 | 180 | 24 | 36 | 3 | 2 | 360

Core • 11-16 • 8 | 36 | 15 | 6 | 9 | 64

Challenge • 19-26 • 90 | 60 | 84 | 120 | 180 | 70 | 160 | 108

Thu • Error Journal Sweep

OUT OF 22

Warm-Up • 1-12 • 24 | 24 | 360 | 6 | 5 | 2 | 32 | 36 | 360 | 10 | 8 | 4

Core • 13-22 • 42 | 9 | 120 | 18 | yes | 68 | 12 | 145 | 18 | yes

Challenge • 23-26 • 100 | 60 | 225 | 45

Fri • Mission 06 Test

OUT OF 20

Warm-Up • 1-4 • 24 | 20 | 26 | 12

Core • 5-20 • 96 | 4 | 25 | 45 | 7 | 20 | 5 | 60 | 144 | 5 | 100 | 77 | 8 |
24 | 7 | 45

Challenge • 21-24 • 320 | 24 | 630 | 35